

A duplication-provenance $\times$ genomic-arrangement framework for paralogous transcription factors in the human immune system

iteration15 (human-only) — paralogous-TF genomics platform

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Abstract

We classify every human paralogous transcription factor (TF) on a 2×2 of **duplication provenance** (2R whole-genome-duplication ohnolog vs. small-scale duplication, SSD) $\times$ **genomic arrangement** (clustered *cis* vs. dispersed), after removing the silent KRAB-ZNF and HOX arrays that confounded earlier analyses. Asking which of the four categories is enriched for cell-type *identity* markers versus cytokine-*response* genes, we find a clean dissociation: enrichment for lineage-identity markers is a specific property of *dispersed* ohnologs and is *erased* by clustering (PBMC odds ratio, OR= 1.79; spleen OR= 2.87; within-ohnolog clustered-vs-dispersed OR $\approx$ 0.33–0.47), whereas cytokine-response enrichment tracks ohnolog *provenance* and *survives* clustering. Dosage analysis shows buffering is not a category-level trait, but specific clustered neighbours diverge: SSD tandems co-amplify (SAND/SP100 family, buffering index B up to 1.84) while a *cis*-ohnolog pair, STAT1/STAT4, is the cleanest buffer ($B = 0.78$, partial $r = -0.41$). We interpret clustering as an evolutionary mechanism that *re-routes* the regulatory potential of a 2R duplicate—from divergent lineage specialisation toward a coordinated, dosage-coupled signalling-effector module, the configuration exemplified by the ETS (ETS1/FLI1/ETS2/ERG) and STAT paralogs.

1 Background and analytical framework

Gene duplication is the raw material of regulatory innovation, but not all duplicates are equivalent. Two axes matter. **Provenance**: a duplicate born in the two rounds of vertebrate whole-genome duplication (2R-WGD, ~ 500 Mya)—an *ohnolog*—is dosage-balanced and preferentially retained, whereas a small-scale duplicate (SSD; tandem or dispersed) is not. **Arrangement**: duplicates that remain in a genomic *cluster* share enhancers and topological context, while *dispersed* copies sit in independent regulatory landscapes. iteration15 crosses these into a single four-category label, `dup_class4`:

$$\{2R \text{ ohnolog}, SSD\} \times \{\text{clustered } cis, \text{dispersed}\} \rightarrow \{\text{dispersed_ohnolog}, \text{clustered_cis_ohnolog}, \text{clustered_SSD_tandem}, \text{dispersed_SSD}\}.$$

A decisive control precedes the analysis: the KRAB-ZNF and HOX arrays—large, silent, tandem-duplicated blocks shown previously to manufacture most of the raw “clustered” enrichment signal—are removed *up front*, so every result below is the KRAB/HOX-controlled effect.

2 Methods (summary)

Human TFs were curated from Lambert et al. (*Cell* 2018); families were defined by shared DNA-binding domain (DBD), with C2H2 zinc-fingers subdivided by HGNC accessory domain (KRAB/SCAN/BTB) and CSD RNA-binders removed. Members were placed on hg38 (GENCODE v45); KRAB-ZNF and HOX were excluded, and survivors were split into genomic clusters (two same-family members adjacent iff no *other* protein-coding gene lies between them) versus singletons. Ohnolog provenance

came from OHNOLOGS v2 plus a general *cis*-ohnolog rescue rule (any cluster containing ≥ 1 known ohnolog confers ohnolog status on all members), which recovers the textbook FLI1/ERG pair with no hand curation. Enrichment was tested per category by one-vs-rest Fisher exact tests (OR, 95% CI, BH-FDR) against the detectable paralogous-TF universe ($\text{catalog} \cap \text{scRNA-seq var_names}$), with pairwise contrasts and a provenance $\times$ arrangement factorial logit. Dosage buffering used the Oesinghaus et al. (2025) human cytokine dictionary: per within-family pair, buffering index $B = \text{Var}(x+y)/(\text{Var } x + \text{Var } y)$ and a common-mode-removed partial correlation, benchmarked against a cross-family null.

3 Part I–II: the four-category catalog

After KRAB/HOX removal, 1,163 paralogous TFs survive, forming 78 genomic clusters (184 clustered members) plus 979 singletons. Following pseudogene/readthrough cleanup, the classified universe is **1,153 TFs** (Table 1). The ETS quartet (ETS1/FLI1/ETS2/ERG) and the STAT family (STAT1/3/4/5A/5B) fall in **clustered_cis_ohnolog**; the chromosome-2 SAND cluster (SP100/110/140/140L), which contains no ohnolog member, remains **clustered_SSD_tandem**.

Table 1: The four duplication categories ($n = 1,153$ classified paralogous TFs).

Category	n	Reference members
dispersed_ohnolog	524	scattered 2R-WGD singletons
dispersed_SSD	450	scattered small-scale duplicates
clustered_cis_ohnolog	106	ETS1/FLI1/ETS2/ERG , STAT1/3/4/5A/5B
clustered_SSD_tandem	73	chr2 SAND: SP100/110/140/140L

4 Part III: identity vs. cytokine-response enrichment

4.1 Cell-type identity markers — a property of *dispersed* ohnologs

In both human PBMC and spleen, **dispersed_ohnolog** is the *only* FDR-significant identity-enriched group, and the two independent marker sets agree (Table 2, Figs. 1–2). Clustering abolishes the effect: within ohnologs, the clustered-vs-dispersed OR is 0.47 (PBMC, $p = 0.008$) and 0.33 (spleen, $p = 0.008$); both clustered categories collapse to the SSD background rate.

Table 2: Identity-marker enrichment (one-vs-rest Fisher). FDR by Benjamini–Hochberg.

Tissue	Category	rate	OR	FDR	
PBMC	dispersed_ohnolog	0.287	1.79	0.0002	enriched $\uparrow$
PBMC	dispersed_SSD	0.192	0.70	0.029	depleted
PBMC	clustered_cis_ohnolog	0.160	0.61	0.119	ns
PBMC	clustered_SSD_tandem	0.164	0.64	0.196	ns
Spleen	dispersed_ohnolog	0.163	2.87	7×10^{-7}	enriched $\uparrow$
Spleen	dispersed_SSD	0.063	0.42	0.0001	depleted
Spleen	clustered_cis_ohnolog	0.061	0.51	0.169	ns
Spleen	clustered_SSD_tandem	0.070	0.60	0.331	ns

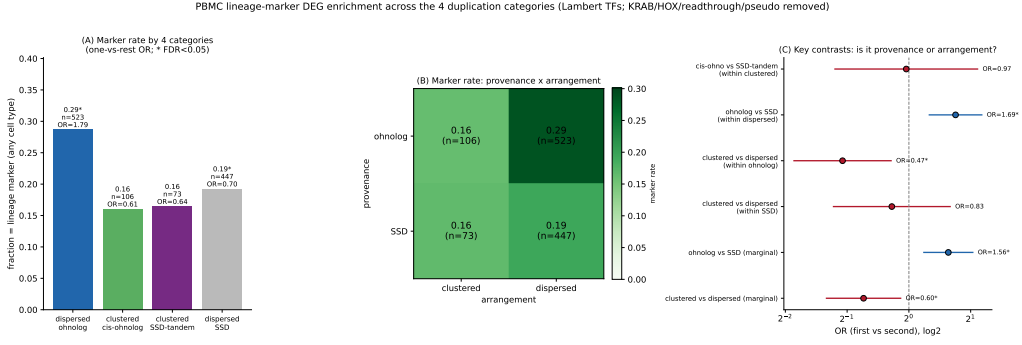


Figure 1: PBMC identity-marker enrichment across the four duplication categories. [dispersed_ohnolog](#) is the single enriched group.

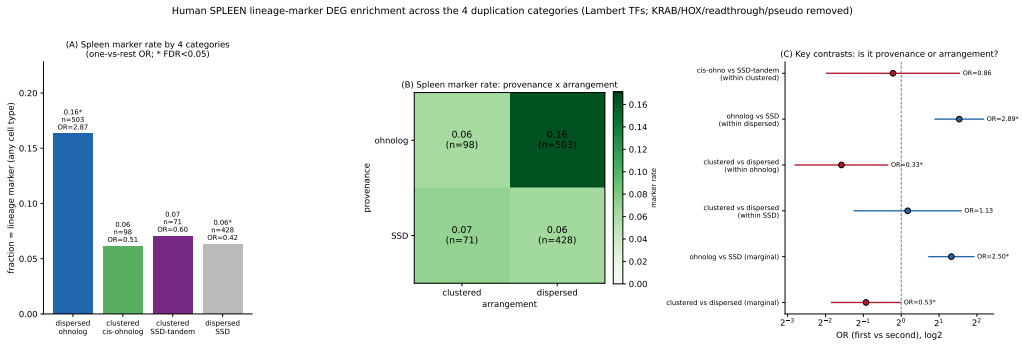


Figure 2: Spleen identity-marker enrichment (independent one-vs-rest markers) reproduces the PBMC pattern: enrichment is exclusive to [dispersed_ohnolog](#).

4.2 Cytokine-response genes — a property of ohnolog *provenance* that survives clustering

The cytokine-response picture dissociates sharply from identity (Table 3, Fig. 3). [dispersed_ohnolog](#) is again most enriched (rate 37.5%, OR= 1.67, FDR= 2.3×10^{-4}), but [clustered_cis_ohnolog](#) *stays high* (rate 29.2%, OR= 0.89, $p = 0.66$, i.e. not depleted), and the within-ohnolog clustered-vs-dispersed contrast is *not* significant ($p = 0.12$)—unlike identity. The provenance $\times$ arrangement factorial confirms the asymmetry: the ohnolog main effect is significant for all three readouts (PBMC adj-OR= 1.69; spleen 2.89; cytokine 1.67), the arrangement main effect and interaction are not.

Table 3: Cytokine-response enrichment (one-vs-rest Fisher).

Category	rate	OR	FDR	
dispersed_ohnolog	0.375	1.67	0.0002	enriched $\uparrow$
clustered_cis_ohnolog	0.292	0.89	0.661	retained (ns)
dispersed_SSD	0.264	0.68	0.007	depleted
clustered_SSD_tandem	0.219	0.59	0.119	ns

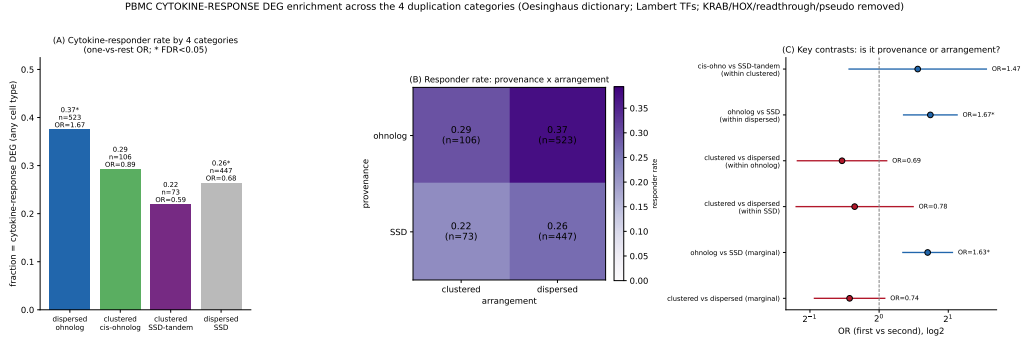


Figure 3: Cytokine-response enrichment. Unlike identity, the clustered *cis*-ohnolog group does not collapse—clustering preserves the cytokine programme.

5 Part IV: dosage buffering by category

Buffering is *not* a category-level trait—every category’s median $B \approx 1.0$, indistinguishable from the cross-family null (Kruskal–Wallis $p = 0.14$; Fig. 4). The signal lives in specific *neighbour* pairs, where the two clustered categories diverge (*cis* vs. SSD-tandem neighbour B contrast $p = 0.038$):

- **SSD-tandem neighbours co-amplify.** The chr2 SAND/SP100 family shows B up to 1.84 (SP100/SP110, partial $r = 0.82$); all six SAND pairs have $B = 1.4$ –1.8 with strong positive partial correlation—redundant gene-dose stacking.
- **A *cis*-ohnolog pair is the cleanest buffer.** STAT1/STAT4 has $B = 0.78$, and common-mode removal flips its correlation negative ($r = -0.27 \rightarrow$ partial $r = -0.41$): active compensation, not shared drive.

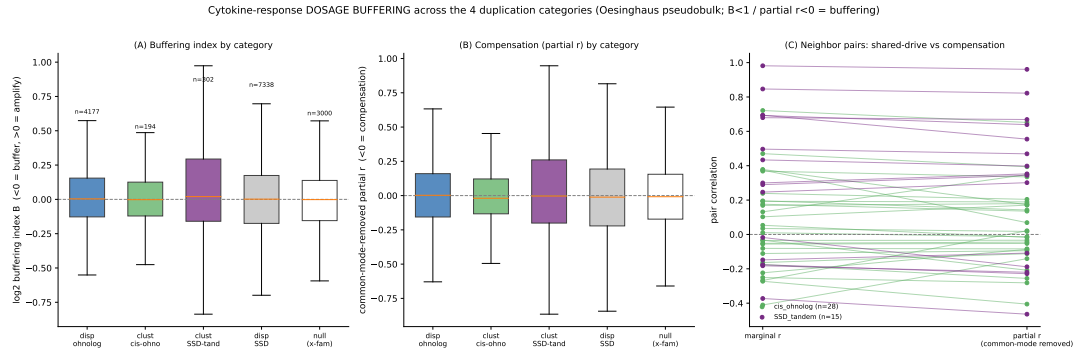


Figure 4: Cytokine-dictionary dosage buffering by category. Category medians sit on the null; divergence is a neighbour-level, pair-specific phenomenon (SAND amplification vs. STAT buffering).

6 Part V: alpha vs. beta ohnolog lineages (Zhu et al. 2026)

Zhu et al. (Nature 2026) report that the jawed-vertebrate second WGD was an **allopolyploidization** fusing two parental lineages, **alpha** and **beta**; alpha-derived ohnologs were $\sim 4\times$ more likely to be retained and, in the *brain*, were more strongly associated with cell-type marker genes than beta. We tested whether this alpha/beta distinction predicts enrichment in the *immune* system.

Using the Marletaz hagfish-paralogue assignments (alpha/beta via chicken orthology, the source Zhu cite), we labelled each ohnolog TF and re-ran the identity- and cytokine-DEG tests (Table 4, Fig. 5).

Of 629 detectable ohnolog-provenance TFs, 451 received a confident lineage call, **343 alpha vs. 108 beta** (3.2:1)—reproducing the alpha-retention bias. Two findings follow. First, **Zhu’s ohnolog > SSD enrichment replicates in immune cells**: both lineages exceed the **dispersed_SSD** reference for identity (alpha 26%, beta 32% vs. 19%) and for cytokine response (36–37% vs. 26%). Second, **the alpha > beta asymmetry does not carry over**: the alpha-vs-beta contrast is non-significant for both readouts (identity OR= 0.74, $p = 0.22$; cytokine OR= 0.96, $p = 0.91$). If anything the roles split by modality—*beta* is the only lineage with a significant one-vs-rest identity enrichment (OR= 1.69, $p = 0.022$), whereas *alpha* carries the cytokine-response signal (OR= 1.36, $p = 0.026$). The robust, transferable claim is therefore ohnolog provenance (> SSD), consistent with Parts III–IV; the brain-specific alpha dominance is not an immune-system property.

Table 4: Alpha vs. beta ohnolog DEG enrichment (human PBMC). Rates are the fraction of each group that is a marker / responder; ORs from two-sided Fisher tests.

Modality	Group	rate	OR	p	
PBMC identity	alpha ($n=343$)	0.262			
PBMC identity	beta ($n=108$)	0.324			
PBMC identity	dispersed_SSD (ref)	0.192			
PBMC identity	alpha vs. beta		0.74	0.219	ns
PBMC identity	beta vs. rest		1.69	0.022	enriched $\uparrow$
Cytokine resp.	alpha ($n=343$)	0.362			
Cytokine resp.	beta ($n=108$)	0.370			
Cytokine resp.	dispersed_SSD (ref)	0.264			
Cytokine resp.	alpha vs. beta		0.96	0.909	ns
Cytokine resp.	alpha vs. rest		1.36	0.026	enriched $\uparrow$

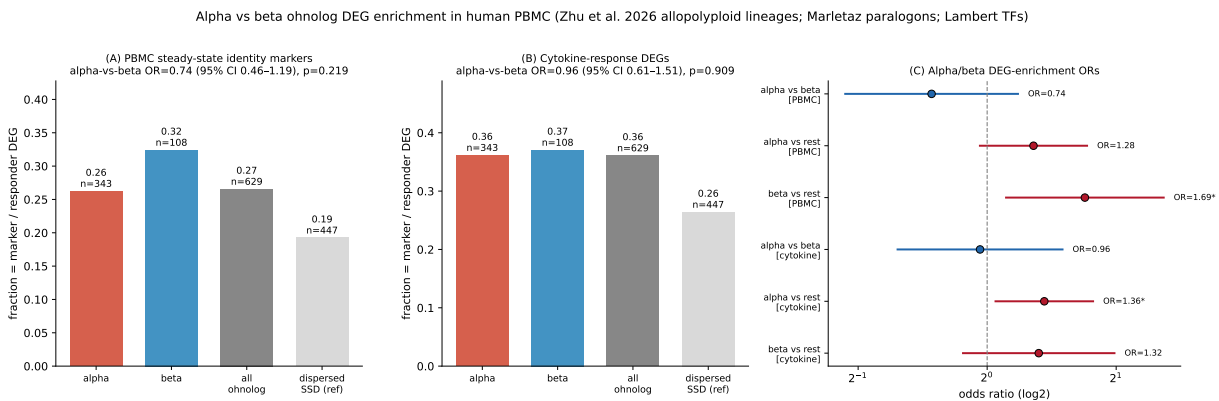


Figure 5: Alpha vs. beta ohnolog DEG enrichment in human PBMC. (A–B) marker/responder rates for alpha, beta, all ohnologs and the dispersed-SSD reference; (C) forest of enrichment ORs. Both lineages beat SSD, but alpha and beta are statistically indistinguishable—the brain alpha-dominance does not transfer, and the immune split is instead beta $\rightarrow$ identity, alpha $\rightarrow$ cytokine.

7 Part VI: monogenic immune-disease (IEI) gene enrichment

If 2R-ohnolog provenance marks dosage-sensitive regulators, those same genes should be over-represented among *monogenic* immune diseases, where a single perturbed copy causes overt disease. We tested this directly against the Genomics England PanelApp curation of inborn errors of immunity (IEI; panel “Primary immunodeficiency or monogenic inflammatory bowel disease” plus “Autoinflammatory disorders”, 378 diagnostic-grade “green” genes), intersected with the 1,153-TF catalog by HGNC identifier. Forty-two paralogous TFs are green IEI genes, and they concentrate sharply by *provenance* (Table 5, Fig. 6).

Table 5: Monogenic-IEI enrichment across the four duplication categories (one-vs-rest Fisher; PanelApp green genes).

Category	IEI rate	OR	p	
dispersed_ohnolog	0.059	3.53	0.0002	enriched $\uparrow$
clustered_cis_ohnolog	0.038	1.04	0.790	background (ns)
clustered_SSD_tandem	0.014	0.35	0.513	ns
dispersed_SSD	0.013	0.25	0.0006	depleted $\downarrow$

The driver is **provenance, not arrangement**. Ohnologs are $\sim 4\times$ more likely than SSDs to be monogenic-IEI genes (OR= 4.34, $p = 1 \times 10^{-4}$), and pre-R1 ancient genes are enriched over younger ones (OR= 3.40, $p = 0.003$), but the buffered/unbuffered axis itself is not significant (clustered vs. dispersed OR= 0.73, $p = 0.67$). The effect is strongest—as the dosage model predicts—for the **monoallelic / dominant** (haploinsufficiency-compatible) green subset, where **dispersed_ohnolog** reaches OR= 4.55 ($p = 0.0004$). Strikingly, the disease reservoir is the *same* **dispersed_ohnolog** category that Part III identified as the cell-type *identity* TFs—the chain closes: the dispersed 2R ohnologs that build immune-cell identity are, when haploinsufficient, the genes that cause monogenic immune disease.

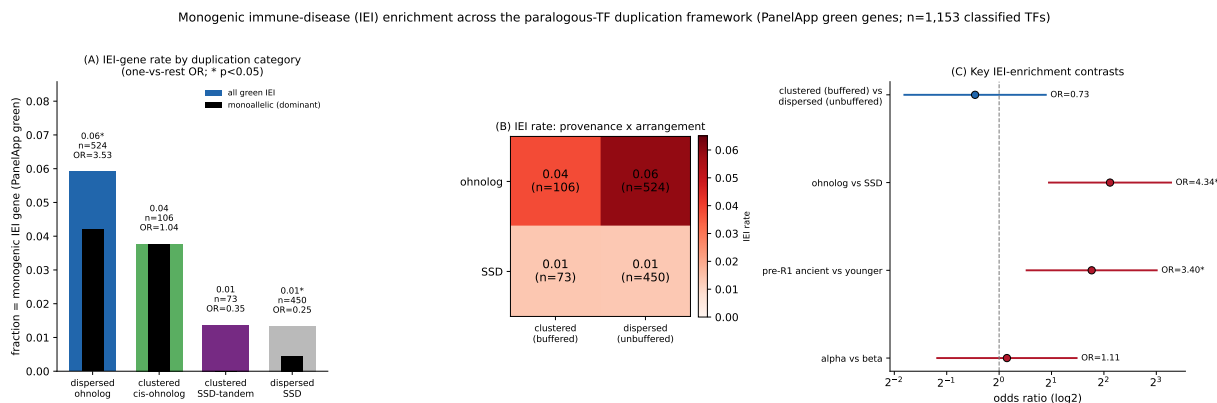


Figure 6: Monogenic-IEI enrichment across the duplication framework. (A) IEI-gene rate by category (black overlay = monoallelic/dominant subset); (B) provenance $\times$ arrangement IEI rate; (C) key contrasts—provenance and ancient origin drive the signal, buffering does not.

The member genes make the pattern concrete. The 31 **dispersed_ohnolog** IEI genes include GATA2, IKZF1, **SPI1**, IRF8, IRF4, **ELF4**, NFKB1/2, REL/RELA/RELB, BACH2, AIRE, FOXN1 and RORC; the four **clustered_cis_ohnolog** IEI genes are the entire STAT cluster (STAT1/3/4/5B); **clustered_SSD_tandem** contributes SP110; and **dispersed_SSD** contributes IRF1, FOXP3, TBX1,

ZBTB24, ZNF341 and ANKZF1.

Relevance to immune disease. Two of three archetypal ETS-pathway disease genes fall in the enriched [dispersed_ohnolog](#) category and are green IEI genes: **SPI1/PU.1** (immunodeficiency) and **ELF4** (autoinflammation). The third, **ETS1/FLI1** (autoimmunity), is the clustered *cis*-ohnolog buffered pair and is *not* a monogenic IEI gene—it is a polygenic lupus GWAS locus instead. This suggests a clean model in which **paralog buffering converts a would-be monogenic dosage gene into a polygenic risk locus**: the autoimmunity pole is common-variant-driven precisely because the neighbouring paralog buffers monogenic penetrance.

8 Part VII: polygenic (GWAS) complex-immune-disease enrichment

Monogenic IEI captures only one end of the genetic-architecture spectrum. The dosage/buffering model makes a sharp prediction at the other end: a paralog *buffered* pair, shielded from monogenic penetrance by its neighbour, should surface instead as a *polygenic* common-variant risk locus. We tested this by curating GWAS genes for five complex immune diseases and re-running the same category enrichment.

Curation. From the NHGRI-EBI GWAS Catalog (full ontology-annotated associations, genome-wide significant at $P \leq 5 \times 10^{-8}$) we extracted nearest/overlapping genes for SLE, rheumatoid arthritis, allergy, asthma and type-1 diabetes, restricted to protein-coding, non-MHC genes: 412 (SLE), 444 (RA), 1,608 (allergy), 639 (asthma) and 271 (T1D) genes respectively. The top loci recover the textbook biology (SLE: STAT4, IRF5, BLK, TNFAIP3, **ETS1**; asthma: IL33, IL1RL1, TSLP, GSDMB; T1D: INS, IL2RA, PTPN22, BACH2), validating the pipeline.

Enrichment and the monogenic→polygenic flip. Of 1,153 paralogous TFs, 245 (21%) are complex-immune-disease GWAS genes. Both ohnolog categories are enriched, but the decisive observation is what happens to the buffered [clustered_cis_ohnolog](#) group (Table 6, Fig. 7): it is the *single most enriched* category for polygenic disease (29%, OR= 1.61, $p = 0.045$) despite sitting at *background* for monogenic IEI (Part VI). Ohnolog provenance again drives the signal (ohnolog vs. SSD OR= 2.06, $p = 2 \times 10^{-6}$), while [dispersed_SSD](#) is depleted (OR= 0.50, $p = 1 \times 10^{-5}$).

Table 6: Monogenic (IEI) vs. polygenic (GWAS, union of 5 diseases) disease-gene rates by category. GWAS enrichment is one-vs-rest Fisher within the catalog.

Category	IEI rate	GWAS rate	GWAS OR	p	
dispersed_ohnolog	0.059	0.260	1.67	0.0004	enriched (both)
clustered_cis_ohnolog	0.038	0.292	1.61	0.0454	polygenic-only ↑
clustered_SSD_tandem	0.014	0.164	0.72	0.375	ns
dispersed_SSD	0.013	0.147	0.50	0.00001	depleted (both)

The ETS pole, mechanistically resolved. The paralogous TFs that are GWAS-associated but *not* monogenic-IEI in the [clustered_cis_ohnolog](#) category are the ETS archetypes themselves: **ETS1**, **FLI1**, **ERG**, together with STAT5A, BATF, EHF and ETV3. Thus ETS1/FLI1/ERG are lupus and allergy GWAS loci yet carry no Mendelian immune disease—exactly the behaviour the buffering model predicts for a common-variant autoimmune-risk locus. Across Parts VI–VII a single axis emerges: **paralog buffering sets genetic architecture**. Unbuffered dispersed

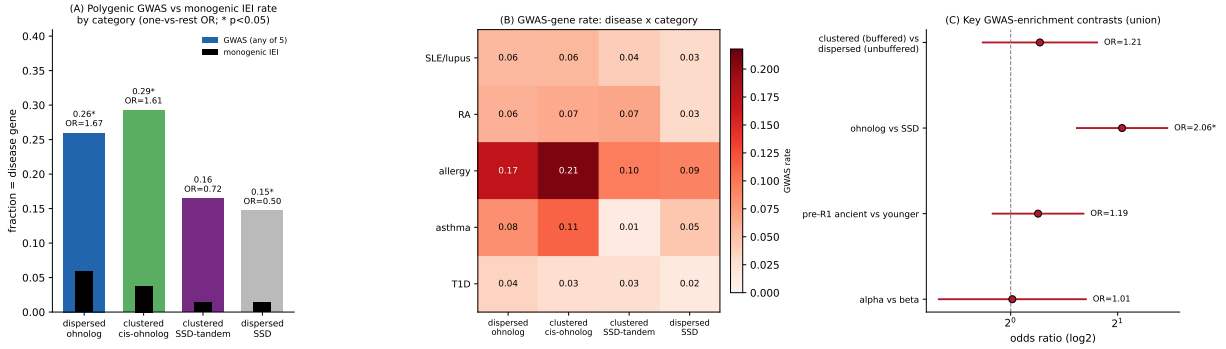


Figure 7: Polygenic GWAS enrichment across the duplication framework. (A) GWAS rate by category with the monogenic-IEI rate overlaid (black)—note `clustered_cis_ohnolog` flips from IEI-background to the top GWAS category; (B) GWAS rate per disease $\times$ category (allergy/asthma drive the clustered-*cis*-ohnolog signal); (C) contrasts—ohnolog provenance drives polygenic risk.

ohnologs (SPI1/PU.1, ELF4) cause *monogenic* immune disease; the buffered clustered *cis*-ohnolog ETS/STAT module instead contributes *polygenic* common-variant risk.

9 Synthesis: what evolutionary advantage might clustering confer?

The central result is a **dissociation**: *provenance* (being a 2R ohnolog) sets a duplicate’s regulatory potential, but *arrangement* (clustered vs. dispersed) decides how that potential is spent. The two classical post-duplication fates—*neofunctionalisation* and *dosage/sub-functional sharing*—map cleanly onto the two arrangements, and clustering appears to carry several distinct selective advantages.

1. Dispersal buys divergence; clustering buys coordination. A dispersed ohnolog sits in its own regulatory landscape and is free to acquire lineage-specific enhancers—hence `dispersed_ohnolog` is the identity-TF reservoir. Clustering removes that freedom: shared enhancers and a shared topological domain constrain the copies to fire together. The advantage of *losing* divergence is **guaranteed co-regulation**: a cluster behaves as one regulatory unit, ensuring its members are co-expressed in the right cell at the right time. This is exactly what we observe—clustered ohnologs lose the (divergence-dependent) identity signal but *retain* the (coordination-dependent) cytokine-response signal.

2. Clustering converts an identity candidate into a signalling effector. The same `clustered_cis_ohnolog` members that are *not* enriched for identity markers *are* enriched (or at least not depleted) for cytokine-response genes. We read this as a functional re-assignment: a 2R ohnolog that might otherwise have become a divergent lineage-determining factor is instead recruited, as a coordinated block, into a fast, signal-responsive effector programme. The ETS (ETS1/FLI1/ETS2/ERG) and STAT paralogs—the engines of antigen-receptor and cytokine signalling—are the archetypes. The selective advantage is a **ready-made, dosage-matched module** that can be deployed coherently downstream of a signal.

3. Clustering enables paralog buffering at dosage-sensitive nodes. True compensatory buffering (STAT1/STAT4: partial $r = -0.41$) is found in a *cis*-ohnolog pair, consistent with the idea that physical proximity plus shared ancestry maintains functional redundancy precisely where dosage matters. Buffering is selectively valuable because it makes the network **robust to perturbation of a single member**: if one paralog is lost or mutated, the neighbour compensates. This is the molecular basis for a paralog-buffered network whose phenotype on perturbation depends on how much redundancy remains.

4. Clustering can also stack dose, not just buffer it. The SSD-tandem SAND/SP100 family shows the opposite neighbour behaviour—co-amplification ($B > 1$). Here the advantage of physical clustering is **additive dosage**: tandem copies of the same response amplify output, the simplest adaptive payoff of a local duplication, with no regulatory rewiring. Thus clustering is not a single strategy; its consequence (buffer vs. amplify) is itself gated by provenance (ohnolog vs. SSD).

In one sentence: a 2R-WGD origin gives a TF pair the *capacity* to become either a divergent identity factor or a coordinated effector, and genomic clustering is the switch—trading the evolvability of dispersal for the robustness and coordination of a co-regulated, dosage-coupled module.

10 Caveats

1. The OR denominator is the detectable paralogous-TF universe, not an expression-matched background; the FDR-backed claims are the [dispersed_ohnolog](#) identity enrichment and the provenance factorial. The within-clustered contrasts rest on small groups ($n \approx 73$ –106) and are trends.
2. Spleen cell-type labels derive from a blood-PBMC Azimuth transfer, so spleen is a reproducibility contrast rather than added cell-type diversity; the cytokine dictionary is also PBMC.
3. Provenance depends on the conservative OHNOLOGS v2; the *cis*-ohnolog rescue fixes only clustered misses (dispersed misses such as FEV stay SSD). The four-category `dup_class4` label—not any per-gene mode—is the unit of analysis.

11 Conclusion

Across human PBMC and spleen identity programmes and a PBMC cytokine dictionary, 2R-WGD ohnolog *provenance* is the determinant of a paralogous TF’s regulatory enrichment, while genomic *arrangement* re-routes it: dispersal toward divergent cell-identity roles, clustering toward coordinated, dosage-coupled cytokine effectors with the capacity for paralog buffering. Clustering’s evolutionary advantage is therefore not a single thing but a trade—coordination, modular deployability and robustness in exchange for the evolvability of independent divergence—and the ETS and STAT paralogs are its clearest immune-system instantiations.

12 Significance: why study this, and relevance to human health and disease

12.1 What iteration15 established

Across this work one coherent idea emerges: **a paralogous transcription factor’s behaviour in the immune system is predicted by how it was born and how it has been kept**. Concretely, (i) the provenance×arrangement classification cleanly separates immune roles—dispersed

ohnologs are the cell-type *identity* reservoir, while clustered *cis*-ohnologs (the ETS and STAT paralogs) are recruited into coordinated cytokine-*effector* modules; (ii) gene-origin dating shows the ohnolog categories are the most ancient ($\sim 72\%$ predate the first WGD) while tandem-SSD pairs are the youngest, the categories differing in origin age at $p = 2 \times 10^{-7}$, and the important singletons BCL6/IRF1/KLF2 are ancient, pre-2R genes *retained single-copy* through both genome duplications; and (iii) layering on the Zhu et al. alpha/beta allopolyploid lineages, the ohnolog-over-SSD enrichment replicates in immune cells but the brain’s alpha-subgenome dominance does not transfer—immune role splits by function, not by ancestral subgenome.

12.2 Why the question is worth studying

1. Dosage is the hidden variable in immune gene regulation. The most reproducible signal throughout is *provenance*: 2R ohnologs were preferentially retained *because their amount matters*, the very property that makes them disease genes—when function depends on dose, one cannot lose, gain, or mistune a copy without consequence. The framework predicts genome-wide which TFs are dosage-critical rather than discovering it one disease at a time.

2. Paralog buffering explains variable penetrance. A mutation in a TF with a redundant, co-expressed neighbour (STAT1/STAT4; ETS1/FLI1) can be silently compensated, while the identical lesion in an unbuffered TF manifests fully—a mechanistic model for the incomplete penetrance and variable expressivity that pervade immune-disease genetics. The age and arrangement results show *where* buffering is even possible: ancient single-copy survivors such as BCL6/IRF1/KLF2 have no paralog to fall back on and are therefore least tolerant of a single hit.

3. Evolution is a natural perturbation screen. Asking which duplicates were kept, in what configuration, and for how long is asking which regulatory architectures selection found indispensable over ~ 500 Myr; ancient retention marks nodes the immune system cannot do without—high-value candidates for both disease causation and therapeutic dependency.

12.3 Relevance to human health and disease

The framework points to a simple organising principle: ETS-family (and more broadly ohnolog) TFs form a dosage-sensitive, paralog-buffered network, and *where* it is perturbed sets the direction of immune disease.

- **One network, the full clinical spectrum.** Three archetypal perturbations of the same ETS/enhancer grammar span the clinical range: **SPI1/PU.1** loss $\rightarrow$ immunodeficiency (CVID); **ELF4** loss $\rightarrow$ autoinflammation (DEX); **ETS1-FLI1** dysregulation $\rightarrow$ autoimmunity (lupus GWAS). The classification predicts why these diverge—it depends on the member, the cell type, and how much paralog redundancy buffers the lesion.
- **Concrete disease genes fall out of the framework.** The ancient single-copy TFs dated here are textbook “unbuffered dosage” immune-disease genes: **IRF1** (Mendelian susceptibility to mycobacterial disease / interferon immunity), **BCL6** (master Tfh/germinal-centre regulator, deregulated in lymphoma), **KLF2** (lymphocyte egress and quiescence); their pre-2R single-copy status *is* the prediction that they are dosage-critical and poorly compensated.
- **Identity vs. effector flags the disease flavour.** Perturbing dispersed ohnologs (identity TFs) tends to reshape which immune cells exist (developmental / immunodeficiency phenotypes); perturbing clustered *cis*-ohnolog effectors (ETS/STAT) tends to derail cytokine responses (autoimmune / autoinflammatory phenotypes).

- **Therapeutic logic.** Buffering predicts both vulnerability and opportunity: where a paralog compensates, one copy can be drugged with tolerable toxicity (or both must be hit for an effect); unbuffered nodes are where dosage-restoring strategies have the largest, clean effect.

In one sentence: by classifying immune TFs by how they were duplicated, how old they are, and how redundant they remain, iteration15 turns “which transcription factor matters” into a predictive statement about dosage sensitivity and buffering—the two parameters that decide whether a genetic lesion produces immunodeficiency, autoinflammation, or autoimmunity, and how it might be treated.

Honesty note: these are association- and evolution-based predictions on a defined paralogous-TF universe, not yet causal tests; the strongest translational payoff is to take the buffered pairs (STAT1/STAT4, ETS1/FLI1) and the unbuffered ancient singletons (IRF1, BCL6, KLF2) into experimental dosage perturbation in primary immune cells.